AEP System Engineering Backlog & Feature Map

Traceability Hierarchy: Requirements (REQ) → Capabilities (CAP) → Features (FEAT) → Tasks (TASK) → Functions (FUNC) → Acceptance Criteria (AC)
Version: v1.1 (Refined & Post-Adversarial Panel Audit) · **Date:** 2026-11-09 · **Branch:** `docs/dual-lane-governance-proposal` · **PR:** #133

1. Adversarial Panel Findings & Architecture Calibrations

> **Adversarial Audit Summary (Planning-Wringer, Paperthin & Uncle Bob):** The adversarial review panel audited the v1.1 backlog against the live repository (`cmd/aep` , `internal/` , `tools/ghg` , `reference/esf`) and surfaced 4 fatal design risks and interface coupling flaws that have been remediated in this refined plan:

Adversarial Finding & Challenge	Ground-Truth Reality & Risk	Applied Backlog Remediation
1. ESF Multi-Backend Gap (Planning-Wringer)	`reference/esf/go.mod:7` hardcodes `CubeSandbox/sdk/go v0.7.3` . Zero native code exists for `libkrun` or `gvisor` . Modifying `reference/esf` is forbidden.	**Descope in-tree ESF modifications:** Cube/KVM is the verified factory path. `libkrun` and `gvisor` are designed as out-of-tree runner adapters behind the `factory.BackendAdapter` seam. Unsupported hosts cleanly quarantine.
2. VirtioFS / Worktree Deadlock (Paperthin Devil's Advocate)	Running 8–10 parallel container lanes mounted to host Git worktrees via VirtioFS on macOS Colima causes severe inode lock contention and `.git/index.lock` race collisions.	**Strict Worktree Isolation:** Each lane operates in an independent branch worktree with isolated index paths (`GIT_INDEX_FILE=/tmp/index`) and dedicated build cache volumes (tmpfs), preventing lock contention.
3. Interface Coupling & SRP Violations (Uncle Bob Clean Architecture)	`lane.Manager` and `factory.Client` were god interfaces bundling process execution, worktree allocation, filesystem fencing, and cryptographic auditing into monolithic structs.	**Interface Segregation (ISP):** Segregated into `WorktreeAllocator` , `ContainerRuntime` , `FenceCoordinator` , `BackendProbe` , and `EgressAuditor` . Pure domain policies separated from disk I/O.
4. Closed-Egress Dependency Catch-22 (Paperthin Adversary)	Enforcing `allow_internet=false` inside Tier-2 sandboxes prevents tasks from downloading missing compiler packages or external test dependencies during builds.	**Pre-Baked Runtime Images & Read-Only Caches:** System dependencies and language modules are pre-seeded in the base lane image or mounted via read-only host volume caches. Egress remains strictly closed.
5. Fatal 8 GB RAM Physical Check Bug (Planning-Wringer)	Backlog AC-07.1.1.3 checked for `< 8 GB available RAM` , which fails on physical 8 GB machines (Intel N100) because OS overhead leaves ~6–7 GB available.	**Check Physical Memory:** AC-07.1.1.3 updated to evaluate *total installed physical RAM* (`Total >= 7.5 GiB`) rather than instantaneous available memory.
6. Redundant Receipt Package (Planning-Wringer)	Backlog invented a redundant `internal/receipt` package duplicating existing `runner.CommandReceipt` in `internal/verifier/runner/runner.go:21-30` .	**SSOT Consolidation:** Merged receipt generation into existing `internal/verifier/runner` primitives, eliminating split-brain receipt structures.

2. Three Deterministic Quality Gates (Uncle Bob's Discipline)

Every PR deliverable must pass three automated, non-negotiable quality gates in CI before merge:

Quality Gate	Threshold & Formula	Tooling & Enforcement	Failure Action
1. CRAP Score Gate (Change Risk Anti-Patterns)	`CRAP(m) = CC(m)^2 * (1 - Cov(m))^3 + CC(m) < 10` Cyclomatic complexity > 8 strictly forbidden.	`gocyclo` + `go test -coverprofile`. Any complex branch must have high test coverage.	PR blocked; function must be split into cohesive sub-functions.
2. Structural DRY Gate (Duplication Elimination)	Token similarity ratio < 0.8 (Minimum 92% non-duplicated code).	Automated AST scanner (`dupl -threshold 35 ./internal/...`). Zero duplicate blocks in decision logic or admission math.	PR blocked; common logic must be consolidated to SSOT helpers.
3. Mutation Kill Gate (Test Kill Efficacy)	Mutation Kill Score > 85% on domain logic. 100% kill on boundary operators (`<`, `<=`, `==`, `!=`).	`go-mutesting` executed on `internal/admission`, `internal/dispatch`, and `internal/policy`.	PR blocked; tests must be enhanced to assert behavior, not line coverage.

3. Feature Map & Dependency Waves (4 Milestones)

Work is organized into 4 non-circular dependency waves, strictly sequenced to enable test-driven progression without mock staleness:

```
WAVE 1: CORE FOUNDATION & SUBSTRATE BOOT (Pure Domain & Primitives)
  ├── PR-01a/b/c: Hardware Profile Domain & Host Prober (internal/profile)
  ├── PR-02a/b:   Policy Tamper Verifier & Env Scrubber (internal/policy)
  └── PR-03a/b:   Memory Relay Client & Go Boot Gate (internal/memoryd, internal/boot)
        │
        ▼
WAVE 2: ADMISSION & INNER-LOOP RALPH FABRIC (Tier 1 Concurrency)
  ├── PR-04a/b:   Pure Admission Engine & FileLock Store (internal/admission)
  ├── PR-05a/b:   Beads Issue Ledger & Dependency DAG Resolver (internal/ledger)
  ├── PR-06a/b:   GitHub Gateway Socket Daemon & Coalescing Cache (internal/gateway)
  └── PR-07a/b/c: Zero-Egress Lane Engine & Container Driver (internal/lane, deploy/compose)
        │
        ▼
WAVE 3: GOVERNANCE & FACTORY MICROVM SANDBOXING (Tier 2 Isolation)
  ├── PR-08a/b:   Pure Dispatch Decision Matrix & Quarantine Audit (internal/dispatch)
  ├── PR-09:      Supervisor Polling Loop & Dispatch Hook (cmd/aep)
  └── PR-10a/b:   ESF Factory Cube/KVM Adapter & Egress Guard (internal/factory)
        │
        ▼
WAVE 4: VERIFICATION, RECEIPTS & SETTLEMENT (Completion & Audit)
  ├── PR-11a/b:   Disposable Sandbox Re-Verification Runner (internal/verifier)
  └── PR-12:      Cryptographic CommandReceipt & Beads Issue Closer (internal/ledger)
```

4. Atomic Work Units & Sizing Matrix (21 PR Slices)

In accordance with Uncle Bob's small-commit discipline, the 11 high-level tasks are broken into 21 bite-sized, atomic deliverables (≤ 300–400 LOC each):

PR Slice	Deliverable Package & Responsibility	Est. LOC	Primary Quality & Test Focus
PR-01a	`internal/profile`: Domain models, presets (`nano-8g` to `workstation-48g`), validation math	~180	Pure unit tests; assert overcommit rejection and physical RAM arithmetic.
PR-01b	`internal/profile`: Hardware probers (`/proc/cpuinfo`, `/dev/kvm`, `sysctl`) behind `HostProbe` seam	~240	Faked OS probes for Linux KVM, macOS VZ, and N100 KVM-free hosts.
PR-01c	`cmd/aep`: Wire `aep profile detect` and `aep profile show` CLI subcommands	~160	Golden output CLI tests; JSON formatting.
PR-02a	`internal/policy`: Environment sanitizer (`ScrubEnv`) and policy token parser	~210	Table tests verifying zero leakage of `GITHUB_TOKEN`, `PAT`, or API secrets.
PR-02b	`internal/policy`: Git HEAD-blob cryptographic tamper verifier (`git rev-parse HEAD`)	~250	Git fixture test; proves uncommitted modifications fail closed.
PR-03a	`internal/memoryd`: Host socket relay client & CLI subprocess bridge	~260	UNIX socket connection resilience; graceful failure on offline daemon.
PR-03b	`internal/boot`: Go-native boot gate orchestrator replacing `boot-gate.sh`	~320	Parity test against shell gate; verifies identical `BOOT-GATE:` output tokens.
PR-04a	`internal/admission`: Pure admission calculation engine (memory budgets & lane limits)	~220	Concurrent race tests with 20 goroutines; mutation score > 85%.
PR-04b	`internal/admission`: File-locked state persistence (`admission.json` using `lock.FileLock`)	~280	Process kill fault-injection; stale lock recovery after 30s.
PR-05a	`internal/ledger`: Beads domain models, blocker DAG resolution, missing-tag policy	~240	Pure graph tests; unclosed blockers defer candidate issues.
PR-05b	`internal/ledger`: `bd` CLI JSON runner adapter with unpinned limit guard (`--limit 0`)	~270	JSON streaming fixture tests; `ErrFetchCeilingExceeded` validation.
PR-06a	`internal/gateway`: Wire protocol, rate limiter, and PR status coalescing cache	~290	Coalescing test: 10 parallel queries collapsed into 1 upstream call.
PR-06b	`internal/gateway`: `aep gwd` UNIX domain socket daemon and host `ghg.py` bridge	~310	Socket client/server IPC tests; branch name scoping security.
PR-07a	`internal/lane`: Worktree allocator & file-lock fence state machine	~280	Isolated worktree path verification; zero collision in parent repo.
PR-07b	`internal/lane`: OCI container runtime driver (Docker/Podman zero-egress runner)	~340	Behavioral container test: verified `ENETUNREACH` and read-only rootfs.
PR-07c	`cmd/aep`: Wire `aep lane claim|release|reap` CLI subcommands	~210	Zombie container reaper test; launcher crash recovery.
PR-08a	`internal/dispatch`: Pure decision engine (evaluating LLD §7 decision matrix)	~320	100% table coverage of every row in §7; quarantine-first on missing KVM.
PR-08b	`internal/dispatch`: Decision audit recorder (`decisions/<ts>-<issue>.json`) & dry-run CLI	~230	Audit persistence; verifies zero silent downgrades to Tier 1.
PR-09	`cmd/aep`: Supervisor daemon integration (polling loop under `--dispatch`)	~260	Idempotent loop test; unassigned work claimed and dispatched cleanly.

PR Slice	Deliverable Package & Responsibility	Est. LOC	Primary Quality & Test Focus
PR-10a	`internal/factory` : Factory config parser, open-egress guard, Cube/KVM adapter	~350	Rejection of `acknowledge_open_egress=true` with exit code 4.
PR-10b	`internal/factory` : In-VM egress probe SHA-256 evidence validator & quarantine hooks	~310	Cryptographic probe validation; corrupted evidence blocks PR creation.
PR-11a	`internal/verifier` : Disposable microVM test runner executing repo build/test scripts	~360	Isolation test: untrusted patch tests execute inside disposable sandbox only.
PR-11b	`internal/verifier` : SHA-256 Merkle tail digest computation and `bd close` settlement guard	~280	Verified receipt required before issue state transitions to closed.

5. Detailed Backlog Specifications (ISP / SRP Refactored)

`CAP-01` **Tier-1 Ralph Container Isolation** `REQ-01.1 · Package: internal/lane`

Executes inner-loop tasks in ephemeral OCI containers with isolated network namespaces, tmpfs `/tmp`, virtiofs git worktrees, and IPC relays.

Segregated Clean Interfaces (ISP):

```
type WorktreeAllocator interface {
  Allocate(ctx context.Context, issueID string) (worktreePath string, err error)
  Free(ctx context.Context, worktreePath string) error
}
type ContainerRuntime interface {
  Start(ctx context.Context, spec ContainerSpec) (containerID string, err error)
  Wait(ctx context.Context, containerID string) (exitCode int, err error)
  Stop(ctx context.Context, containerID string) error
}
type LaneManager struct {
  allocator WorktreeAllocator
  runtime ContainerRuntime
  fences FenceStore
}
```

Behavioral Acceptance Criteria (TDD):

- **AC-01.1 (Confinement):** GIVEN a spawned container, WHEN executing any outbound socket call, THEN the kernel returns `ENETUNREACH` or `EPERM` immediately.
- **AC-01.2 (Lock Free):** GIVEN 4 concurrent lanes, WHEN git commits execute simultaneously, THEN isolated index files prevent `.git/index.lock` collisions.
- **AC-01.3 (Reap Safety):** GIVEN a launcher process killed with `SIGKILL` while the container runs, WHEN `aep lane reap` runs, THEN it marks the lane active and refuses to delete state until the container stops.

CAP-02 Tier-2 Local-First Sandbox Factory

REQ-01.2 · Package: `internal/factory`

Executes untrusted dependencies, multi-repo changes, and CVE remediation in hardware microVMs with measured egress.

Segregated Clean Interfaces (ISP):

```
type BackendProbe interface {
  Probe(ctx context.Context) (SandboxBackend, error)
}
type PipelineRunner interface {
  Submit(ctx context.Context, req RunRequest) (handle RunHandle, err error)
  Status(ctx context.Context, handle RunHandle) (RunStatus, error)
}
type EgressAuditor interface {
  VerifyProbeDigest(ctx context.Context, evidencePath string) (bool, error)
}
```

Behavioral Acceptance Criteria (TDD):

- **AC-02.1 (Hardening Guard):** GIVEN `factory.toml` with `hardening.acknowledge_open_egress = true`, WHEN submitted, THEN runner rejects with exit 4.
- **AC-02.2 (Backend Resolution):** GIVEN Linux with `/dev/kvm`, THEN selects `cube-kvm`; GIVEN host lacking hardware virtualization, THEN selects `gvisor` adapter or safely quarantines.
- **AC-02.3 (Dependency Pre-Baking):** GIVEN closed egress (`allow_internet=false`), WHEN build scripts run, THEN required dependencies resolve from pre-seeded image layers or read-only volume mounts without external network access.

CAP-07 Multi-Hardware Profile Registry & Admission

REQ-04.1 · Package: `internal/profile` & `internal/admission`

Enforces hardware budget scaling from 2 vCPU / 8 GB (Intel N100) up to 12 vCPU / 48 GB (Apple M3 Max) with dual-constraint admission.

Pure Domain Logic Decoupled from I/O:

```
type Usage struct {
  ActiveLanes int
  AllocatedRAM int
}
// Pure calculation without disk locking or syscalls:
func CalculateAdmission(u Usage, p Profile, req ReservationReq) (*Reservation, error)
```

Behavioral Acceptance Criteria (TDD):

- **AC-07.1 (Physical Memory Floor):** GIVEN a host with `TotalRAM >= 7.5 GiB` (physical 8 GB), WHEN `aep profile detect` runs, THEN it successfully assigns preset `nano-8g` regardless of ephemeral available RAM.
- **AC-07.2 (Dual-Constraint Concurrency):** GIVEN an active memory budget of 4.0 GB, WHEN two 2.0 GB lanes are active, THEN a 3rd lane request receives `ErrNoCapacity` even if the lane count ceiling is not reached.